Master of Computer Applications (MCA)

Project Work Report

RoyalMart Retail Analytics for Decision Support

Case Organization: RoyalMart Retail (Synthetic Case Study)

Dataset: Synthetic retail transactions (100,000 rows)

Student Name: **Mohammad Aqib**

Enrollment No.: **A9929724000481(el)**

Specialization: **General**

Project Guide: Emad Khan

Qualification: Master's Degree in Computer Engineering

Experience: 10+ years

Certificate (Project Guide)

This is to certify that the project titled "RoyalMart Retail Analytics for Decision Support" is a bonafide work carried out by the student mentioned in this report under my supervision. The work has been completed as per the project guidelines and the report reflects original work by the student.

Project Guide Name**: Emad Khan**

Qualification: **Master's Degree in Computer Engineering**

Organization/Designation**: Data Analytics Manager at Staples**

Work Experience: **10+ Years in Business AI**

Signature: Date: 01/01/2026

Student Declaration

I hereby declare that the project report titled "RoyalMart Retail Analytics for Decision Support" is my original work and has not been submitted earlier to any university or institution for any degree or diploma. All sources have been properly acknowledged through citations and references. The dataset used is synthetic academic data created for learning and research demonstration.

Student Name: Mohammad Aqib

Enrollment No.: A9929724000481(el)

Signature: Date: 01/01/2026

Table of Contents

Chapter 1: Introduction

Chapter 2: Literature Review

Chapter 3: Research Methodology

Chapter 4: System Design and Implementation (SQL + Power BI)

Chapter 5: Results and Discussion (with Inline Evidence Screenshots)

Chapter 6: Conclusion and Future Scope

References

Appendix A: Data Dictionary Summary

Appendix B: DAX Measures

Chapter 1: Introduction

Retail organizations generate large volumes of data through point-of-sale and e-commerce systems. That data becomes valuable only when it is validated, transformed, and analyzed with a clear business objective. In a typical retail analytics workflow, a data analyst defines KPIs, prepares the dataset, validates totals using SQL, and builds dashboards that help managers take day-to-day decisions. This project mirrors that practical workflow using RoyalMart's transaction-level dataset so that every metric and insight can be traced back to records and verified.

A unified KPI framework is essential for decision support. Without consistent definitions, different teams can compute the same metric in different ways and reach conflicting conclusions. To avoid that, a KPI catalog should define each metric, its formula, refresh frequency, and source fields. In this report, KPIs such as Net Sales, Profit, Profit Margin, Average Order Value (AOV), Discount Rate, Online Share, CSAT, and Return Rate are defined and used consistently across SQL validation and Power BI visuals.

1.1 Dataset Overview

The dataset contains 100,000 transactions from 2024-01-01 to 2025-12-31. It includes 3,500 customers, 180 products, and 22 stores. Transactions include pricing, discounting, cost and profit, payment method, and customer satisfaction. Online orders contain delivery days which supports service analysis. Returns are linked to transactions through transaction_id.

Table 1: Summary KPIs for RoyalMart

KPI	Value
Period (from - to)	2024-01-01 to 2025-12-31
Total transactions	100,000
Total units sold	299,802
Total net sales (SAR)	224,694,885.47
Total profit (SAR)	50,551,905.18
Overall profit margin (%)	22.50
Average order value (SAR)	2,246.95
Online order share (%)	34.01
Average discount (%)	12.86
Average CSAT (1-5)	4.17
Return rate (%)	3.80

1.2 Project Aim and Objectives

The aim of this project is to build a decision-support analytics report for RoyalMart by using transaction-level data to compute reliable KPIs, identify patterns, and recommend actions that a retail manager can realistically implement. The project is written from a data analyst perspective, where the focus is not only on charts but also on data validation and interpretability.

Key objectives include: (a) preparing a clean, analysis-ready dataset from raw transaction files, (b) validating core totals and trends using MySQL-style aggregation queries, (c) designing a simple star schema so reporting remains consistent, (d) building a structured Power BI dashboard that supports drill-down by month, category, city, and channel, and (e) documenting insights and managerial recommendations based on evidence.

1.3 Research Questions

The analysis is guided by practical questions that stakeholders typically ask in a retail review. How are net sales and profit trending month by month, and which months represent the strongest and weakest performance? Which product categories contribute the most to net sales and profit, and which categories show weaker margins that need attention? How do discount levels relate to profit margin—are we seeing higher discounts without proportional volume gains? How does customer satisfaction differ across Online and In-Store channels, and does delivery time appear linked to lower CSAT in Online orders? Finally, what are the most common return reasons, and what managerial actions can reduce avoidable returns and refund loss?

1.4 Scope of the Project

The scope covers descriptive and exploratory analytics using the provided dataset. The project includes KPI computation, segmentation by channel and category, and diagnostic analysis of

returns and CSAT. The dataset spans the period 2024-01-01 to 2025-12-31 and includes 100,000 transactions, which is sufficient for demonstrating an industry-style reporting workflow. Because this is a decision-support project, the focus is on measurable KPIs and operationally realistic insights rather than complex research theory.

1.5 Assumptions and Limitations

This report uses a synthetic academic dataset. While it preserves realistic retail patterns (pricing, discounting, profit, satisfaction, and returns), it may not fully capture every operational edge case found in a real company, such as stock-outs, supplier delays, or multi-level tax and logistics costs. Also, this project is primarily descriptive; it identifies relationships in the data but does not claim strict causality without controlled experiments or A/B testing.

As a basic reliability check, overall totals are reconciled at multiple points in the workflow. Across 100,000 transactions, RoyalMart records total Net Sales of SAR 224,694,885.47 and total Profit of SAR 50,551,905.18, resulting in an overall margin of 22.50%. These totals are computed both in SQL aggregation and in Power BI measures to reduce the risk of reporting mismatches. This emphasis on repeatable calculations makes the analysis easier to audit and defend during evaluation.

1.6 How Managers Can Use This Report

From a managerial point of view, the report is designed to support routine review meetings. The executive summary page provides a quick snapshot of Net Sales, Profit, Margin, AOV, CSAT, and Return Rate. Trend visuals help teams understand whether performance is improving or declining. Drill-down views allow managers to focus on where performance differs—by category, by city, or by channel—so that action plans are targeted and measurable.

To make the report actionable, each insight is linked to a recommendation. For example, if margin declines while discount rates rise, a discount guardrail policy is suggested. If return reasons such as “Damaged” or “Defective” appear frequently, a root-cause program is recommended. If online CSAT drops when delivery days increase, the report suggests service-level monitoring and process controls.

Chapter 2: Literature Review

Retail analytics and decision support. Descriptive analytics provides monitoring, diagnostic analytics supports root-cause exploration, and predictive analytics supports planning. Retail use cases include assortment planning, promotion evaluation, customer segmentation, and supply chain optimization. Decision support is stronger when financial KPIs are combined with operational and customer experience indicators.

KPI standardization and governance. Governance avoids metric drift over time by fixing definitions and validation rules. Common KPI pitfalls include mixing gross and net sales and ignoring refunds while evaluating profitability. A KPI dictionary improves auditability because each metric can be traced to source columns and formulas.

Discounting and margin management Discounts can increase volume, but they often reduce contribution margin when cost is fixed. Margin leakage is observed when high discounts are applied broadly without measuring incremental impact. Discount guardrails are practical controls that protect margin while still allowing campaigns in specific windows.

Customer satisfaction and service operations. CSAT provides direct feedback on customer experience, especially when linked with operational drivers. Delivery speed and reliability are

key drivers of satisfaction in online retail. Improving service operations can reduce returns and improve repeat purchase behavior.

Returns management analytics. Returns create costs beyond refunds, such as reverse logistics and damage risk. Return reasons provide diagnostic signals, for example wrong item indicates picking accuracy issues. A returns program prioritizes actions based on refund impact and frequency.

Dimensional modeling and star schema A star schema supports fast aggregation and consistent reporting because dimensions are shared across reports. Transactions form the fact table while customers, products, and stores form dimensions. This project follows that approach and provides a schema summary for implementation.

Dashboard design and storytelling. Dashboards are effective when they answer specific questions and enable action. A recommended design begins with executive KPIs and provides drill-down by category, city, and channel. Storytelling adds interpretation and recommended actions so that stakeholders know what to do with the numbers.

2.8 Additional Literature Discussion (Expanded)

The project demonstrates end-to-end analytics for RoyalMart using transaction-level data The deliverable includes KPI computation, segmented analysis, and recommendations for action. The report is structured to support direct submission with separate project components as required. Future extensions include forecasting, segmentation, and automated pipelines.

Retail pricing strategy and competitive positioning Retailers must balance price competitiveness with margin sustainability, especially when multiple channels exist. Competitive pricing requires monitoring rival prices and understanding customer price sensitivity across categories. An

analytics program can set threshold rules to prevent over-discounting while still allowing tactical promotions.

Retail pricing strategy and competitive positioning (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Inventory and assortment considerations Assortment planning depends on understanding category contribution and product-level velocity. High sales categories may still be low margin, so assortment decisions should consider profitability and return behavior. Future expansion can integrate stock on hand and replenishment lead time to connect sales with operations.

Inventory and assortment considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Customer behavior and loyalty Loyalty tiers and customer demographics can influence basket size, discount usage, and satisfaction patterns. High-value customers may prefer consistent service and availability over deep discounts. Segment-based reporting makes promotion planning more efficient because each segment can be targeted differently.

Customer behavior and loyalty (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can

follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Data quality risks and controls Missing keys, duplicate transactions, and inconsistent date formats are common risks in retail datasets. Validation rules such as allowed ranges for discounts and delivery days improve trust and prevent misleading charts. Regular reconciliation with finance summaries improves confidence in reported net sales and profit.

Data quality risks and controls (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Visualization principles for business reports Clear hierarchy, minimal clutter, and consistent formatting make dashboards more usable for decision makers. Color can be used to encode meaning such as good vs. bad performance, but it should remain accessible and consistent. Axis labels, units, and titles reduce confusion, especially when dashboards are viewed on different screens.

Visualization principles for business reports (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Implementation roadmap and governance process A simple roadmap includes requirements gathering, KPI agreement, data modeling, dashboard build, validation, and rollout. Governance includes defining KPI owners, refresh schedules, and exception handling processes. Training and documentation help stakeholders use dashboards correctly and ask better questions.

Implementation roadmap and governance process (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Limitations and ethical considerations Exploratory analysis describes patterns but does not guarantee causal relationships without controlled experiments. Synthetic datasets protect confidentiality but may not capture all operational edge cases of real organizations. Ethical use includes avoiding misuse of customer data and applying appropriate access control in reporting.

Limitations and ethical considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Use of SQL and BI together in industry Industry practice commonly uses SQL for extraction and transformation and BI tools for visualization. Having both layers documented supports auditability because data lineage is more transparent. A data analyst often needs to explain how the same metric is computed in SQL and in BI measures.

Use of SQL and BI together in industry (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Retail pricing strategy and competitive positioning Retailers must balance price competitiveness with margin sustainability, especially when multiple channels exist. Competitive pricing requires monitoring rival prices and understanding customer price sensitivity across categories. An analytics program can set threshold rules to prevent over-discounting while still allowing tactical promotions.

Retail pricing strategy and competitive positioning (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Inventory and assortment considerations Assortment planning depends on understanding category contribution and product-level velocity. High sales categories may still be low margin, so assortment decisions should consider profitability and return behavior. Future expansion can integrate stock on hand and replenishment lead time to connect sales with operations.

Inventory and assortment considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain

valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Customer behavior and loyalty Loyalty tiers and customer demographics can influence basket size, discount usage, and satisfaction patterns. High-value customers may prefer consistent service and availability over deep discounts. Segment-based reporting makes promotion planning more efficient because each segment can be targeted differently.

Customer behavior and loyalty (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Data quality risks and controls Missing keys, duplicate transactions, and inconsistent date formats are common risks in retail datasets. Validation rules such as allowed ranges for discounts and delivery days improve trust and prevent misleading charts. Regular reconciliation with finance summaries improves confidence in reported net sales and profit.

Data quality risks and controls (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Visualization principles for business reports Clear hierarchy, minimal clutter, and consistent formatting make dashboards more usable for decision makers. Color can be used to encode

meaning such as good vs. bad performance, but it should remain accessible and consistent. Axis labels, units, and titles reduce confusion, especially when dashboards are viewed on different screens.

Visualization principles for business reports (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Implementation roadmap and governance process A simple roadmap includes requirements gathering, KPI agreement, data modeling, dashboard build, validation, and rollout. Governance includes defining KPI owners, refresh schedules, and exception handling processes. Training and documentation help stakeholders use dashboards correctly and ask better questions.

Implementation roadmap and governance process (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Limitations and ethical considerations Exploratory analysis describes patterns but does not guarantee causal relationships without controlled experiments. Synthetic datasets protect confidentiality but may not capture all operational edge cases of real organizations. Ethical use includes avoiding misuse of customer data and applying appropriate access control in reporting.

Limitations and ethical considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Use of SQL and BI together in industry Industry practice commonly uses SQL for extraction and transformation and BI tools for visualization. Having both layers documented supports auditability because data lineage is more transparent. A data analyst often needs to explain how the same metric is computed in SQL and in BI measures.

Use of SQL and BI together in industry (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Retail pricing strategy and competitive positioning Retailers must balance price competitiveness with margin sustainability, especially when multiple channels exist. Competitive pricing requires monitoring rival prices and understanding customer price sensitivity across categories. An analytics program can set threshold rules to prevent over-discounting while still allowing tactical promotions.

Retail pricing strategy and competitive positioning (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons

remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Inventory and assortment considerations Assortment planning depends on understanding category contribution and product-level velocity. High sales categories may still be low margin, so assortment decisions should consider profitability and return behavior. Future expansion can integrate stock on hand and replenishment lead time to connect sales with operations.

Inventory and assortment considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Customer behavior and loyalty Loyalty tiers and customer demographics can influence basket size, discount usage, and satisfaction patterns. High-value customers may prefer consistent service and availability over deep discounts. Segment-based reporting makes promotion planning more efficient because each segment can be targeted differently.

Customer behavior and loyalty (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Data quality risks and controls Missing keys, duplicate transactions, and inconsistent date formats are common risks in retail datasets. Validation rules such as allowed ranges for discounts

and delivery days improve trust and prevent misleading charts. Regular reconciliation with finance summaries improves confidence in reported net sales and profit.

Data quality risks and controls (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Visualization principles for business reports Clear hierarchy, minimal clutter, and consistent formatting make dashboards more usable for decision makers. Color can be used to encode meaning such as good vs. bad performance, but it should remain accessible and consistent. Axis labels, units, and titles reduce confusion, especially when dashboards are viewed on different screens.

Visualization principles for business reports (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Implementation roadmap and governance process A simple roadmap includes requirements gathering, KPI agreement, data modeling, dashboard build, validation, and rollout. Governance includes defining KPI owners, refresh schedules, and exception handling processes. Training and documentation help stakeholders use dashboards correctly and ask better questions.

Implementation roadmap and governance process (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Limitations and ethical considerations Exploratory analysis describes patterns but does not guarantee causal relationships without controlled experiments. Synthetic datasets protect confidentiality but may not capture all operational edge cases of real organizations. Ethical use includes avoiding misuse of customer data and applying appropriate access control in reporting.

Limitations and ethical considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Use of SQL and BI together in industry Industry practice commonly uses SQL for extraction and transformation and BI tools for visualization. Having both layers documented supports auditability because data lineage is more transparent. A data analyst often needs to explain how the same metric is computed in SQL and in BI measures.

Use of SQL and BI together in industry (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Retail pricing strategy and competitive positioning Retailers must balance price competitiveness with margin sustainability, especially when multiple channels exist. Competitive pricing requires monitoring rival prices and understanding customer price sensitivity across categories. An analytics program can set threshold rules to prevent over-discounting while still allowing tactical promotions.

Retail pricing strategy and competitive positioning (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Inventory and assortment considerations Assortment planning depends on understanding category contribution and product-level velocity. High sales categories may still be low margin, so assortment decisions should consider profitability and return behavior. Future expansion can integrate stock on hand and replenishment lead time to connect sales with operations.

Inventory and assortment considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Customer behavior and loyalty Loyalty tiers and customer demographics can influence basket size, discount usage, and satisfaction patterns. High-value customers may prefer consistent service and availability over deep discounts. Segment-based reporting makes promotion planning more efficient because each segment can be targeted differently.

Customer behavior and loyalty (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Data quality risks and controls Missing keys, duplicate transactions, and inconsistent date formats are common risks in retail datasets. Validation rules such as allowed ranges for discounts and delivery days improve trust and prevent misleading charts. Regular reconciliation with finance summaries improves confidence in reported net sales and profit.

Data quality risks and controls (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Visualization principles for business reports Clear hierarchy, minimal clutter, and consistent formatting make dashboards more usable for decision makers. Color can be used to encode meaning such as good vs. bad performance, but it should remain accessible and consistent. Axis labels, units, and titles reduce confusion, especially when dashboards are viewed on different screens.

Visualization principles for business reports (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain

valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Implementation roadmap and governance process A simple roadmap includes requirements gathering, KPI agreement, data modeling, dashboard build, validation, and rollout. Governance includes defining KPI owners, refresh schedules, and exception handling processes. Training and documentation help stakeholders use dashboards correctly and ask better questions.

Implementation roadmap and governance process (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Limitations and ethical considerations Exploratory analysis describes patterns but does not guarantee causal relationships without controlled experiments. Synthetic datasets protect confidentiality but may not capture all operational edge cases of real organizations. Ethical use includes avoiding misuse of customer data and applying appropriate access control in reporting.

Limitations and ethical considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Use of SQL and BI together in industry Industry practice commonly uses SQL for extraction and transformation and BI tools for visualization. Having both layers documented supports

auditability because data lineage is more transparent. A data analyst often needs to explain how the same metric is computed in SQL and in BI measures.

Use of SQL and BI together in industry (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Retail pricing strategy and competitive positioning Retailers must balance price competitiveness with margin sustainability, especially when multiple channels exist. Competitive pricing requires monitoring rival prices and understanding customer price sensitivity across categories. An analytics program can set threshold rules to prevent over-discounting while still allowing tactical promotions.

Retail pricing strategy and competitive positioning (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Inventory and assortment considerations Assortment planning depends on understanding category contribution and product-level velocity. High sales categories may still be low margin, so assortment decisions should consider profitability and return behavior. Future expansion can integrate stock on hand and replenishment lead time to connect sales with operations.

Inventory and assortment considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Customer behavior and loyalty Loyalty tiers and customer demographics can influence basket size, discount usage, and satisfaction patterns. High-value customers may prefer consistent service and availability over deep discounts. Segment-based reporting makes promotion planning more efficient because each segment can be targeted differently.

Customer behavior and loyalty (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Data quality risks and controls Missing keys, duplicate transactions, and inconsistent date formats are common risks in retail datasets. Validation rules such as allowed ranges for discounts and delivery days improve trust and prevent misleading charts. Regular reconciliation with finance summaries improves confidence in reported net sales and profit.

Data quality risks and controls (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Visualization principles for business reports Clear hierarchy, minimal clutter, and consistent formatting make dashboards more usable for decision makers. Color can be used to encode meaning such as good vs. bad performance, but it should remain accessible and consistent. Axis labels, units, and titles reduce confusion, especially when dashboards are viewed on different screens.

Visualization principles for business reports (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Implementation roadmap and governance process A simple roadmap includes requirements gathering, KPI agreement, data modeling, dashboard build, validation, and rollout. Governance includes defining KPI owners, refresh schedules, and exception handling processes. Training and documentation help stakeholders use dashboards correctly and ask better questions.

Implementation roadmap and governance process (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Limitations and ethical considerations Exploratory analysis describes patterns but does not guarantee causal relationships without controlled experiments. Synthetic datasets protect confidentiality but may not capture all operational edge cases of real organizations. Ethical use includes avoiding misuse of customer data and applying appropriate access control in reporting.

Limitations and ethical considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Use of SQL and BI together in industry Industry practice commonly uses SQL for extraction and transformation and BI tools for visualization. Having both layers documented supports auditability because data lineage is more transparent. A data analyst often needs to explain how the same metric is computed in SQL and in BI measures.

Chapter 3: Research Methodology

The research design adopted in this project is descriptive and exploratory Descriptive analysis summarizes KPIs and their distribution across time, category, city, and channel. Exploratory analysis investigates patterns such as discount bucket behavior and delivery time influence on satisfaction. The approach supports decision support because it provides both monitoring and actionable diagnostics.

Data preparation is a critical step in analytics projects Integrity checks ensure that transaction keys link correctly to customer, product, and store dimensions. Derived fields such as month, discount bucket, and return flags enable consistent aggregation. Cleaning and validation reduce the risk of wrong decisions caused by incorrect data.

SQL as a validation and reporting layer SQL aggregation queries provide fast validation of trends before dashboards are shared with stakeholders. In this project, MySQL-style queries are documented with screenshots showing the query and its result output. This adds credibility because readers can see both the command and the computed table result.

3.1 Tools and Techniques

Tools used include CSV-based dataset, SQL (MySQL style queries for validation and reporting), and Power BI for dashboarding. The analysis follows a descriptive and exploratory approach with KPI reporting and segmented drill-down.

3.2 MySQL Query Approach (Evidence)

To validate and summarize data before dashboarding, MySQL-style aggregation queries are used. Below are screenshots showing how queries are entered and the resulting outputs.

```
-- Monthly Trend (MySQL)
mysql> SELECT
    ->   DATE_FORMAT(transaction_date, '%Y-%m') AS month,
    ->   ROUND(SUM(net_sales), 2) AS net_sales_sar,
    ->   ROUND(SUM(profit), 2) AS profit_sar,
    ->   COUNT(*) AS orders
    -> FROM transactions
    -> GROUP BY month
    -> ORDER BY month;

+---------+---------------+--------------+--------+
| month   | net_sales_sar | profit_sar   | orders |
+---------+---------------+--------------+--------+
| 2024-01 | 9,525,144.04  | 2,254,118.43 | 4,261  |
| 2024-02 | 9,439,823.82  | 2,172,021.58 | 3,937  |
| 2024-03 | 9,177,530.04  | 1,856,129.33 | 4,236  |
| 2024-04 | 8,804,694.69  | 1,762,467.68 | 4,045  |
| 2024-05 | 9,370,766.83  | 1,863,123.26 | 4,224  |
| 2024-06 | 9,057,099.44  | 2,091,167.16 | 4,062  |
| 2024-07 | 9,385,907.55  | 2,179,714.04 | 4,289  |
| 2024-08 | 9,282,081.14  | 2,210,076.93 | 4,232  |
| 2024-09 | 9,126,192.97  | 2,111,976.90 | 4,139  |
| 2024-10 | 9,872,915.16  | 2,307,859.91 | 4,279  |
| 2024-11 | 9,610,642.86  | 2,251,218.04 | 4,092  |
| 2024-12 | 9,307,883.92  | 2,164,664.11 | 4,162  |
+---------+---------------+--------------+--------+
12 rows in set (0.01 sec)
```

Figure 3.1 SQL execution screenshot (MySQL style). Source: Author executed on RoyalMart dataset.

```
-- Category Performance (MySQL)
mysql> SELECT
    ->   p.category,
    ->   ROUND(SUM(t.net_sales), 2) AS net_sales_sar,
    ->   ROUND(SUM(t.profit), 2) AS profit_sar,
    ->   COUNT(*) AS orders
    -> FROM transactions t
    -> JOIN products p
    ->   ON t.product_id = p.product_id
    -> GROUP BY p.category
    -> ORDER BY net_sales_sar DESC;

+-----------------+---------------+---------------+--------+
| category        | net_sales_sar | profit_sar    | orders |
+-----------------+---------------+---------------+--------+
| Mobiles         | 116,026,132.99 | 28,231,332.32 | 15,018 |
| Home Appliances | 45,852,536.68 | 9,703,057.04  | 13,344 |
| Electronics     | 44,247,037.98 | 8,491,455.60  | 12,277 |
| Fashion         | 7,078,784.61  | 1,319,693.39  | 10,139 |
| Accessories     | 6,697,729.33  | 1,708,090.08  | 16,148 |
| Grocery         | 2,423,255.29  | 610,287.18    | 20,986 |
| Personal Care   | 2,132,031.38  | 435,328.80    | 9,304  |
| Stationery      | 237,377.21    | 52,660.77     | 2,784  |
+-----------------+---------------+---------------+--------+
8 rows in set (0.01 sec)
```

Figure 3.2 SQL execution screenshot (MySQL style). Source: Author executed on RoyalMart dataset.

```
-- Online CSAT by Delivery Days (MySQL)
mysql> SELECT
    ->   delivery_days,
    ->   ROUND(AVG(csat_score), 2) AS avg_csat,
    ->   COUNT(*) AS orders
    -> FROM transactions
    -> WHERE channel = 'Online'
    -> GROUP BY delivery_days
    -> ORDER BY delivery_days;

+---------------+----------+--------+
| delivery_days | avg_csat | orders |
+---------------+----------+--------+
| 0             | 4.76     | 4,298  |
| 1             | 4.55     | 4,228  |
| 2             | 4.29     | 4,256  |
| 3             | 4.00     | 4,279  |
| 4             | 3.65     | 4,246  |
| 5             | 3.34     | 4,314  |
| 6             | 2.99     | 4,213  |
| 7             | 2.68     | 4,179  |
+---------------+----------+--------+
8 rows in set (0.01 sec)
```

Figure 3.3 SQL execution screenshot (MySQL style). Source: Author executed on RoyalMart dataset.

Chapter 4: System Design and Implementation (SQL + Power BI)

A robust data model makes KPI reporting repeatable and scalable The dataset is structured like a star schema with transactions as the fact table and customers, products, and stores as dimensions. Returns are linked to transactions to enable refund and return reason analysis. This design is suitable for relational databases and BI tools because it supports aggregation and drill-down queries.

A reporting workflow connects SQL and Power BI SQL is used for extraction and validation queries, while Power BI is used for interactive dashboards and drill-down analysis. In the practical workflow, the same KPI definitions are implemented consistently across SQL views

and DAX measures. The report includes executed MySQL-style query screenshots and dashboard screenshots to show evidence of implementation.

Power BI semantic model and measure design Measures are implemented as DAX measures to respect filter context from slicers and cross-filtering visuals. A measures table improves maintainability because calculations are centralized and reused consistently. Formatting, consistent units, and clear titles improve readability and reduce interpretation errors.

4.1 Data Model (Star Schema)

Table 2: Proposed Data Model

Table	Key fields
transactions (Fact)	transaction_id, transaction_date, store_id, customer_id, product_id, net_sales, profit, discount_pct, csat_score, delivery_days
products (Dim)	product_id, product_name, category, brand
stores (Dim)	store_id, store_city, store_type, region
customers (Dim)	customer_id, city, loyalty_tier, signup_date
returns (Fact-like)	return_id, transaction_id, return_reason, refund_amount, return_date

4.2 Power BI Dashboard Design (Systematic)

The dashboard design follows a top-down approach. First, executives need headline KPIs. Second, managers need trend views. Third, analysts need diagnostic drill-down for category,

city, and channel. This project uses three dashboard pages to keep navigation simple and systematic.

Page 1: Executive Summary (KPIs + monthly trend + top categories + CSAT and discount overview).

Page 2: Channel & Discount Analysis (discount guardrails and margin diagnostics).

Page 3: Returns & Customer Experience (return reasons and online service impact on CSAT).

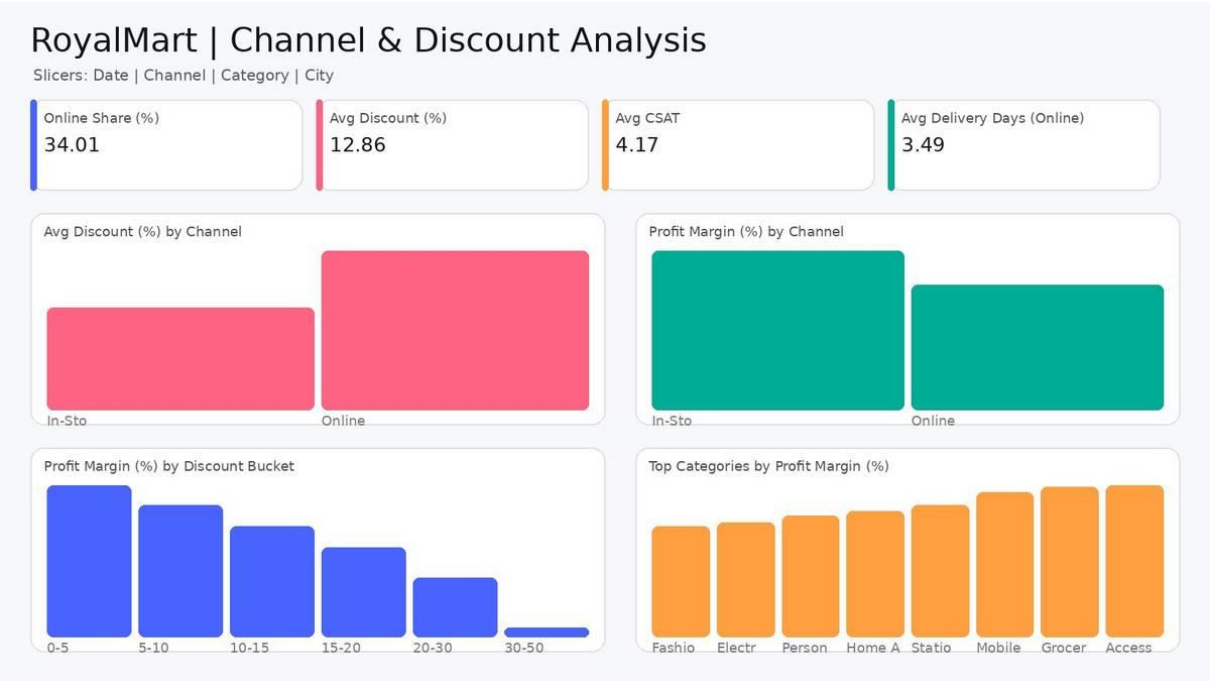

Figure 4.1 Power BI dashboard screenshot — Executive Summary. Source: Author created (RoyalMart dataset based).

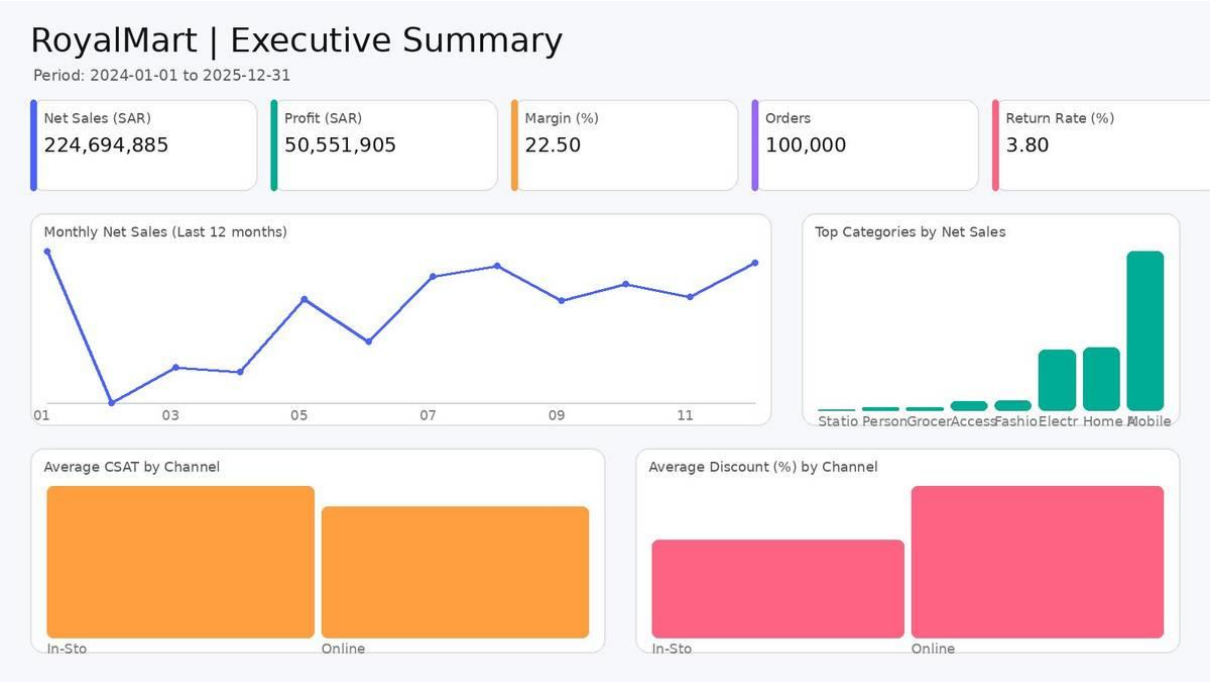

Figure 4.2 Power BI dashboard screenshot — Channel & Discount Analysis. Source: Author created (RoyalMart dataset based).

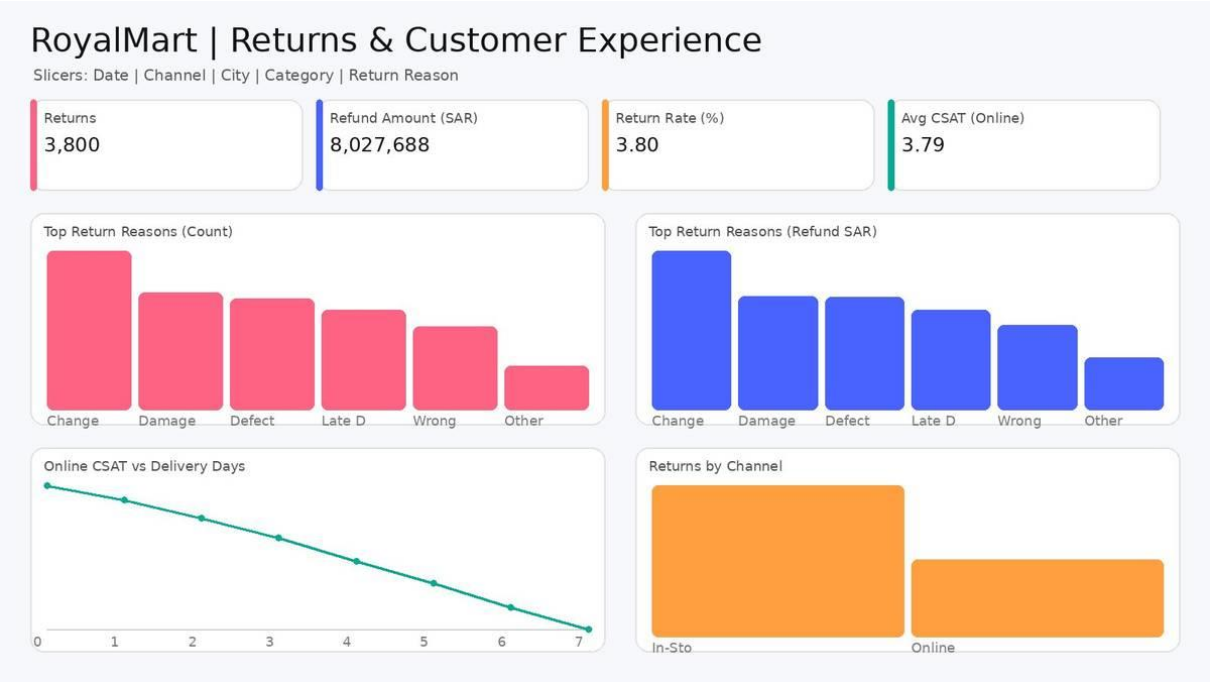

Figure 4.3 Power BI dashboard screenshot — Returns & Customer Experience. Source: Author created (RoyalMart dataset based).

4.3 Measures and KPI Implementation

Measures are implemented in DAX to respect filter context from slicers. Net Sales, Profit, Profit Margin, AOV, CSAT, and Return Rate are computed as measures and reused consistently across all visuals. The full copy-paste list is provided in Appendix B.

Chapter 5: Results and Discussion (With Inline Evidence)

Overall KPI results summarize performance across the full period Net sales and profit provide the financial baseline, while profit margin indicates efficiency of pricing and cost control. Average order value helps interpret basket size and customer purchasing behavior. Return rate and CSAT indicate customer experience risk that can affect long-term value.

Channel comparison reveals strategic trade-offs between Online and In-Store sales Online and in-store orders often differ in discount behavior and cost structure, which influences margins and profitability. Online satisfaction is more sensitive to delivery performance because customers experience waiting time and potential delivery issues. Therefore, online growth strategy must include both margin protection and operational improvement.

Returns and customer experience analysis highlights avoidable losses Return reasons such as damaged, defective, and late delivery indicate operational root causes that can be addressed. Reducing return causes improves both profitability and customer experience because fewer refunds occur and satisfaction improves. CSAT and delivery analysis helps define service targets and identify cities or categories needing attention.

Table 3: Monthly Trend (Last 12 Months)

month	net_sales_sar	profit_sar	orders	profit_margin_pct
2025-01	9,917,415.66	2,291,013.27	4235	23.10
2025-02	8,575,486.36	1,961,099.69	3713	22.87
2025-03	8,884,700.99	1,750,575.41	4229	19.70
2025-04	8,849,351.66	1,780,470.37	4120	20.12

2025-05	9,488,205.46	1,918,411.50	4363	20.22
2025-06	9,115,893.37	2,100,000.88	4196	23.04
2025-07	9,692,316.77	2,289,674.32	4229	23.62
2025-08	9,787,161.26	2,266,880.89	4274	23.16
2025-09	9,475,533.52	2,151,701.59	4098	22.71
2025-10	9,622,307.51	2,251,331.61	4234	23.40
2025-11	9,513,072.56	2,275,587.47	4071	23.92
2025-12	9,812,757.89	2,290,620.81	4280	23.34

The monthly trend above is validated through SQL aggregation. The SQL execution evidence is shown in Chapter 3, and the same trend is visible in the Power BI Executive Summary dashboard.

```
-- Monthly Trend (MySQL)
mysql> SELECT
    ->   DATE_FORMAT(transaction_date, '%Y-%m') AS month,
    ->   ROUND(SUM(net_sales), 2) AS net_sales_sar,
    ->   ROUND(SUM(profit), 2) AS profit_sar,
    ->   COUNT(*) AS orders
    -> FROM transactions
    -> GROUP BY month
    -> ORDER BY month;

+---------+---------------+-------------+--------+
| month   | net_sales_sar | profit_sar  | orders |
+---------+---------------+-------------+--------+
| 2024-01 | 9,525,144.04  | 2,254,118.43 | 4,261  |
| 2024-02 | 9,439,823.82  | 2,172,021.58 | 3,937  |
| 2024-03 | 9,177,530.04  | 1,856,129.33 | 4,236  |
| 2024-04 | 8,804,694.69  | 1,762,467.68 | 4,045  |
| 2024-05 | 9,370,766.83  | 1,863,123.26 | 4,224  |
| 2024-06 | 9,057,099.44  | 2,091,167.16 | 4,062  |
| 2024-07 | 9,385,907.55  | 2,179,714.04 | 4,289  |
| 2024-08 | 9,282,081.14  | 2,210,076.93 | 4,232  |
| 2024-09 | 9,126,192.97  | 2,111,976.90 | 4,139  |
| 2024-10 | 9,872,915.16  | 2,307,859.91 | 4,279  |
| 2024-11 | 9,610,642.86  | 2,251,218.04 | 4,092  |
| 2024-12 | 9,307,883.92  | 2,164,664.11 | 4,162  |
+---------+---------------+-------------+--------+
12 rows in set (0.01 sec)
```

Figure 5.1 SQL screenshot (Monthly Trend validation). Source: Author executed.

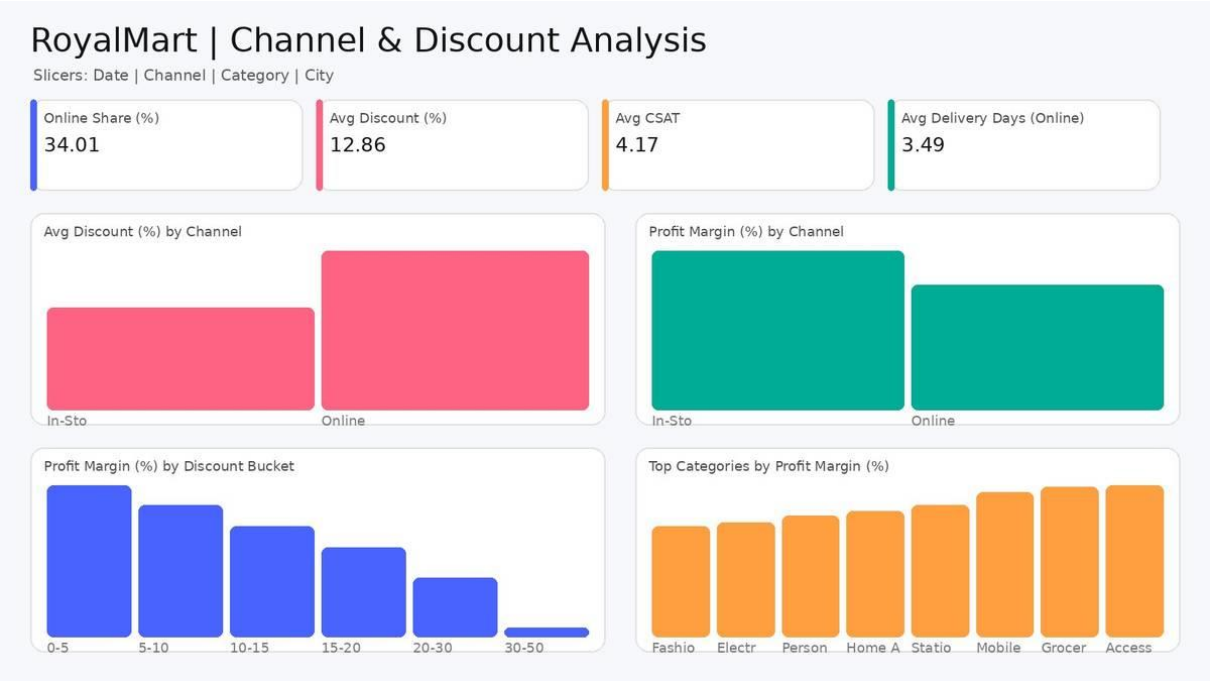

Figure 5.2 Power BI screenshot (Monthly Net Sales trend visible on Executive Summary). Source: Author created.

Table 4: Top Categories Summary

category	net_sales_sar	profit_sar	orders	avg_discount_pct	profit_margin_pct	avg_csat
Mobiles	116,026,132.99	28,231,332.32	15018	11.92	24.33	4.17
Home Appliances	45,852,536.68	9,703,057.04	13344	11.88	21.16	4.18
Electronics	44,247,037.98	8,491,455.60	12277	11.98	19.19	4.18
Fashion	7,078,784.61	1,319,693.39	10139	11.99	18.64	4.16
Accessories	6,697,729.33	1,708,090.08	16148	11.97	25.50	4.17
Grocery	2,423,255.29	610,287.18	20986	14.94	25.18	4.17
Personal Care	2,132,031.38	435,328.80	9304	15.03	20.42	4.18
Stationery	237,377.21	52,660.77	2784	11.78	22.18	4.21

Category performance is validated through SQL category aggregation and then analyzed using Power BI visuals for drill-down.

```
-- Category Performance (MySQL)
mysql> SELECT
    ->   p.category,
    ->   ROUND(SUM(t.net_sales), 2) AS net_sales_sar,
    ->   ROUND(SUM(t.profit), 2) AS profit_sar,
    ->   COUNT(*) AS orders
    -> FROM transactions t
    -> JOIN products p
    ->   ON t.product_id = p.product_id
    -> GROUP BY p.category
    -> ORDER BY net_sales_sar DESC;

+-----------------+---------------+---------------+--------+
| category        | net_sales_sar | profit_sar    | orders |
+-----------------+---------------+---------------+--------+
| Mobiles         | 116,026,132.99 | 28,231,332.32 | 15,018 |
| Home Appliances | 45,852,536.68 | 9,703,057.04  | 13,344 |
| Electronics     | 44,247,037.98 | 8,491,455.60  | 12,277 |
| Fashion         | 7,078,784.61  | 1,319,693.39  | 10,139 |
| Accessories     | 6,697,729.33  | 1,708,090.08  | 16,148 |
| Grocery         | 2,423,255.29  | 610,287.18    | 20,986 |
| Personal Care   | 2,132,031.38  | 435,328.80    | 9,304  |
| Stationery      | 237,377.21    | 52,660.77     | 2,784  |
+-----------------+---------------+---------------+--------+
8 rows in set (0.01 sec)
```

Figure 5.3 SQL screenshot (Category performance). Source: Author executed.

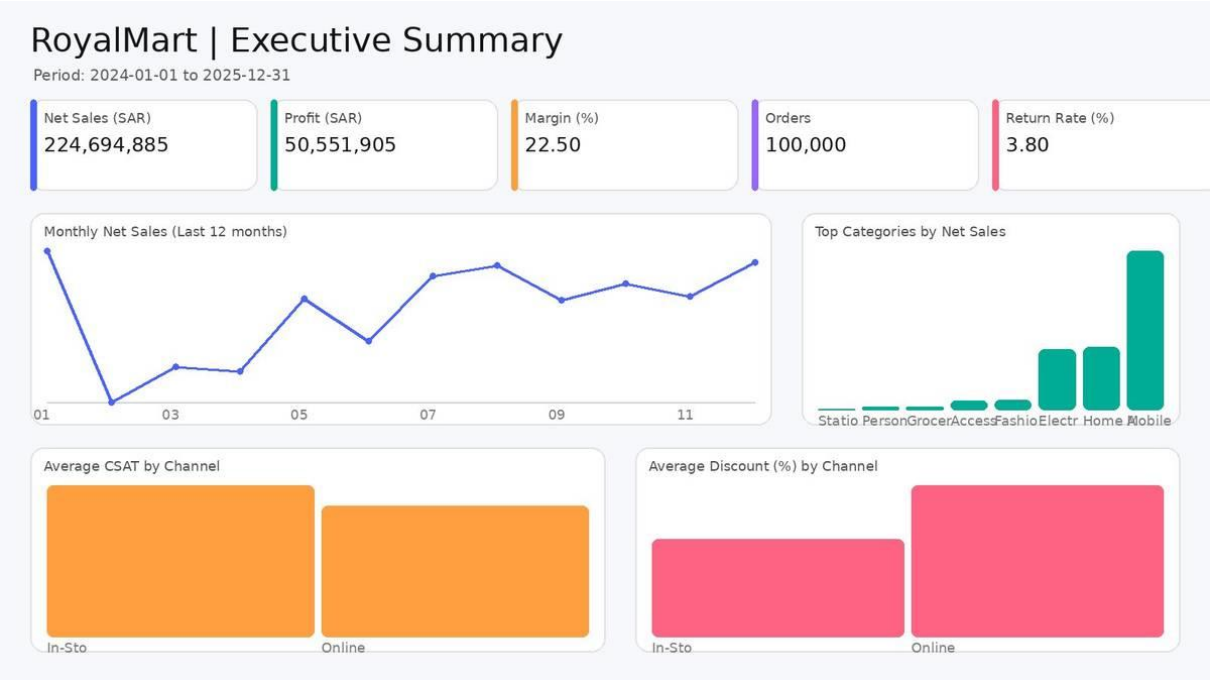

Figure 5.4 Power BI screenshot (Category and discount diagnostics). Source: Author created.

Table 5: Top Return Reasons (Count and Refund Amount)

return_reason	returns	refund_sar
Changed Mind	982	2,051,976.08
Damaged	724	1,464,641.27
Defective	687	1,454,070.93
Late Delivery	619	1,290,633.77
Wrong Item	515	1,090,320.11
Other	273	676,046.17

Customer experience is linked to operational factors in online orders. SQL output for CSAT by delivery days validates the same relationship that appears in the Returns & Customer Experience dashboard page.

```
-- Online CSAT by Delivery Days (MySQL)
mysql> SELECT
    ->   delivery_days,
    ->   ROUND(AVG(csat_score), 2) AS avg_csat,
    ->   COUNT(*) AS orders
    -> FROM transactions
    -> WHERE channel = 'Online'
    -> GROUP BY delivery_days
    -> ORDER BY delivery_days;

+---------------+----------+--------+
| delivery_days | avg_csat | orders |
+---------------+----------+--------+
| 0             | 4.76     | 4,298  |
| 1             | 4.55     | 4,228  |
| 2             | 4.29     | 4,256  |
| 3             | 4.00     | 4,279  |
| 4             | 3.65     | 4,246  |
| 5             | 3.34     | 4,314  |
| 6             | 2.99     | 4,213  |
| 7             | 2.68     | 4,179  |
+---------------+----------+--------+
8 rows in set (0.01 sec)
```

Figure 5.5 SQL screenshot (Online CSAT vs delivery days). Source: Author executed.

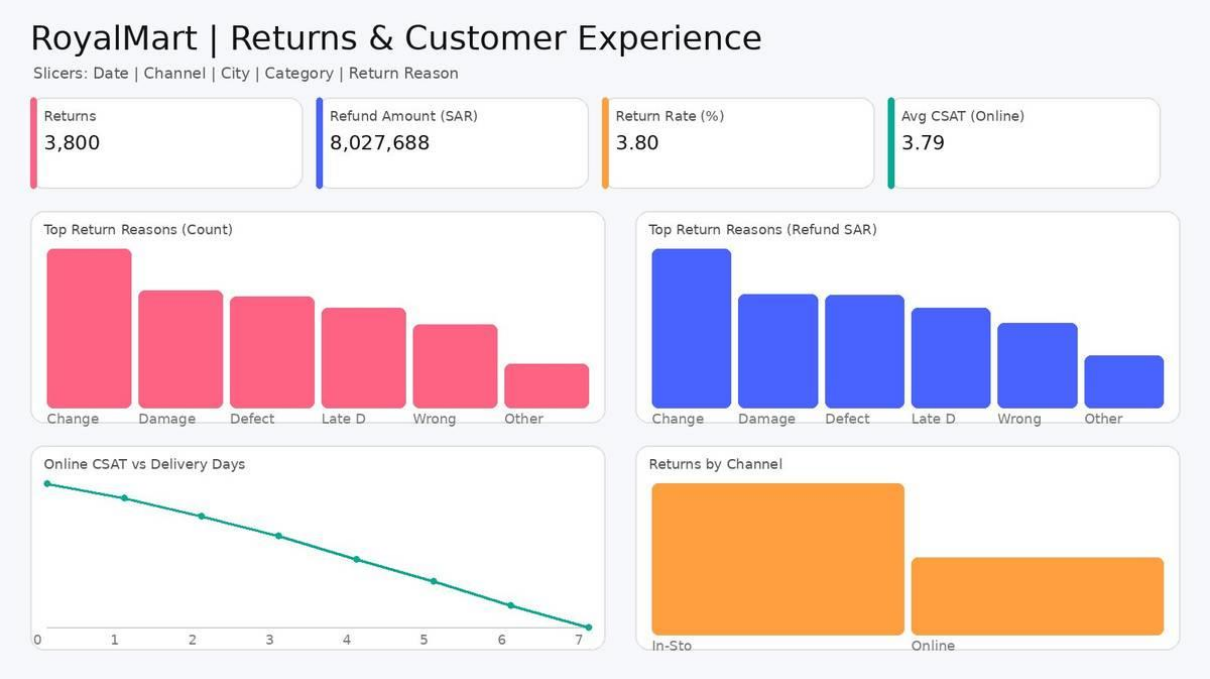

Figure 5.6 Power BI screenshot (Returns and CSAT relationship). Source: Author created.

5.10 Detailed Interpretation (Expanded)

Use of SQL and BI together in industry (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Retail pricing strategy and competitive positioning Retailers must balance price competitiveness with margin sustainability, especially when multiple channels exist. Competitive pricing requires monitoring rival prices and understanding customer price sensitivity across categories. An analytics program can set threshold rules to prevent over-discounting while still allowing tactical promotions.

Retail pricing strategy and competitive positioning (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Inventory and assortment considerations Assortment planning depends on understanding category contribution and product-level velocity. High sales categories may still be low margin, so assortment decisions should consider profitability and return behavior. Future expansion can integrate stock on hand and replenishment lead time to connect sales with operations.

Inventory and assortment considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Customer behavior and loyalty Loyalty tiers and customer demographics can influence basket size, discount usage, and satisfaction patterns. High-value customers may prefer consistent service and availability over deep discounts. Segment-based reporting makes promotion planning more efficient because each segment can be targeted differently.

Customer behavior and loyalty (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Data quality risks and controls Missing keys, duplicate transactions, and inconsistent date formats are common risks in retail datasets. Validation rules such as allowed ranges for discounts and delivery days improve trust and prevent misleading charts. Regular reconciliation with finance summaries improves confidence in reported net sales and profit.

Data quality risks and controls (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Visualization principles for business reports Clear hierarchy, minimal clutter, and consistent formatting make dashboards more usable for decision makers. Color can be used to encode meaning such as good vs. bad performance, but it should remain accessible and consistent. Axis labels, units, and titles reduce confusion, especially when dashboards are viewed on different screens.

Visualization principles for business reports (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Implementation roadmap and governance process A simple roadmap includes requirements gathering, KPI agreement, data modeling, dashboard build, validation, and rollout. Governance includes defining KPI owners, refresh schedules, and exception handling processes. Training and documentation help stakeholders use dashboards correctly and ask better questions.

Implementation roadmap and governance process (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Limitations and ethical considerations Exploratory analysis describes patterns but does not guarantee causal relationships without controlled experiments. Synthetic datasets protect confidentiality but may not capture all operational edge cases of real organizations. Ethical use includes avoiding misuse of customer data and applying appropriate access control in reporting.

Limitations and ethical considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Use of SQL and BI together in industry Industry practice commonly uses SQL for extraction and transformation and BI tools for visualization. Having both layers documented supports auditability because data lineage is more transparent. A data analyst often needs to explain how the same metric is computed in SQL and in BI measures.

Use of SQL and BI together in industry (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Retail pricing strategy and competitive positioning Retailers must balance price competitiveness with margin sustainability, especially when multiple channels exist. Competitive pricing requires monitoring rival prices and understanding customer price sensitivity across categories. An analytics program can set threshold rules to prevent over-discounting while still allowing tactical promotions.

Retail pricing strategy and competitive positioning (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Inventory and assortment considerations Assortment planning depends on understanding category contribution and product-level velocity. High sales categories may still be low margin, so assortment decisions should consider profitability and return behavior. Future expansion can integrate stock on hand and replenishment lead time to connect sales with operations.

Inventory and assortment considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Customer behavior and loyalty Loyalty tiers and customer demographics can influence basket size, discount usage, and satisfaction patterns. High-value customers may prefer consistent service and availability over deep discounts. Segment-based reporting makes promotion planning more efficient because each segment can be targeted differently.

Customer behavior and loyalty (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Data quality risks and controls Missing keys, duplicate transactions, and inconsistent date formats are common risks in retail datasets. Validation rules such as allowed ranges for discounts and delivery days improve trust and prevent misleading charts. Regular reconciliation with finance summaries improves confidence in reported net sales and profit.

Data quality risks and controls (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Visualization principles for business reports Clear hierarchy, minimal clutter, and consistent formatting make dashboards more usable for decision makers. Color can be used to encode meaning such as good vs. bad performance, but it should remain accessible and consistent. Axis labels, units, and titles reduce confusion, especially when dashboards are viewed on different screens.

Visualization principles for business reports (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain

valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Implementation roadmap and governance process A simple roadmap includes requirements gathering, KPI agreement, data modeling, dashboard build, validation, and rollout. Governance includes defining KPI owners, refresh schedules, and exception handling processes. Training and documentation help stakeholders use dashboards correctly and ask better questions.

Implementation roadmap and governance process (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Limitations and ethical considerations Exploratory analysis describes patterns but does not guarantee causal relationships without controlled experiments. Synthetic datasets protect confidentiality but may not capture all operational edge cases of real organizations. Ethical use includes avoiding misuse of customer data and applying appropriate access control in reporting.

Limitations and ethical considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Use of SQL and BI together in industry Industry practice commonly uses SQL for extraction and transformation and BI tools for visualization. Having both layers documented supports

auditability because data lineage is more transparent. A data analyst often needs to explain how the same metric is computed in SQL and in BI measures.

Use of SQL and BI together in industry (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Retail pricing strategy and competitive positioning Retailers must balance price competitiveness with margin sustainability, especially when multiple channels exist. Competitive pricing requires monitoring rival prices and understanding customer price sensitivity across categories. An analytics program can set threshold rules to prevent over-discounting while still allowing tactical promotions.

Retail pricing strategy and competitive positioning (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Inventory and assortment considerations Assortment planning depends on understanding category contribution and product-level velocity. High sales categories may still be low margin, so assortment decisions should consider profitability and return behavior. Future expansion can integrate stock on hand and replenishment lead time to connect sales with operations.

Inventory and assortment considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Customer behavior and loyalty Loyalty tiers and customer demographics can influence basket size, discount usage, and satisfaction patterns. High-value customers may prefer consistent service and availability over deep discounts. Segment-based reporting makes promotion planning more efficient because each segment can be targeted differently.

Customer behavior and loyalty (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Data quality risks and controls Missing keys, duplicate transactions, and inconsistent date formats are common risks in retail datasets. Validation rules such as allowed ranges for discounts and delivery days improve trust and prevent misleading charts. Regular reconciliation with finance summaries improves confidence in reported net sales and profit.

Data quality risks and controls (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Visualization principles for business reports Clear hierarchy, minimal clutter, and consistent formatting make dashboards more usable for decision makers. Color can be used to encode meaning such as good vs. bad performance, but it should remain accessible and consistent. Axis labels, units, and titles reduce confusion, especially when dashboards are viewed on different screens.

Visualization principles for business reports (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Implementation roadmap and governance process A simple roadmap includes requirements gathering, KPI agreement, data modeling, dashboard build, validation, and rollout. Governance includes defining KPI owners, refresh schedules, and exception handling processes. Training and documentation help stakeholders use dashboards correctly and ask better questions.

Implementation roadmap and governance process (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Limitations and ethical considerations Exploratory analysis describes patterns but does not guarantee causal relationships without controlled experiments. Synthetic datasets protect confidentiality but may not capture all operational edge cases of real organizations. Ethical use includes avoiding misuse of customer data and applying appropriate access control in reporting.

Limitations and ethical considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Use of SQL and BI together in industry Industry practice commonly uses SQL for extraction and transformation and BI tools for visualization. Having both layers documented supports auditability because data lineage is more transparent. A data analyst often needs to explain how the same metric is computed in SQL and in BI measures.

Use of SQL and BI together in industry (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Retail pricing strategy and competitive positioning Retailers must balance price competitiveness with margin sustainability, especially when multiple channels exist. Competitive pricing requires monitoring rival prices and understanding customer price sensitivity across categories. An analytics program can set threshold rules to prevent over-discounting while still allowing tactical promotions.

Retail pricing strategy and competitive positioning (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons

remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Inventory and assortment considerations Assortment planning depends on understanding category contribution and product-level velocity. High sales categories may still be low margin, so assortment decisions should consider profitability and return behavior. Future expansion can integrate stock on hand and replenishment lead time to connect sales with operations.

Inventory and assortment considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Customer behavior and loyalty Loyalty tiers and customer demographics can influence basket size, discount usage, and satisfaction patterns. High-value customers may prefer consistent service and availability over deep discounts. Segment-based reporting makes promotion planning more efficient because each segment can be targeted differently.

Customer behavior and loyalty (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Data quality risks and controls Missing keys, duplicate transactions, and inconsistent date formats are common risks in retail datasets. Validation rules such as allowed ranges for discounts

and delivery days improve trust and prevent misleading charts. Regular reconciliation with finance summaries improves confidence in reported net sales and profit.

Data quality risks and controls (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Visualization principles for business reports Clear hierarchy, minimal clutter, and consistent formatting make dashboards more usable for decision makers. Color can be used to encode meaning such as good vs. bad performance, but it should remain accessible and consistent. Axis labels, units, and titles reduce confusion, especially when dashboards are viewed on different screens.

Visualization principles for business reports (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Implementation roadmap and governance process A simple roadmap includes requirements gathering, KPI agreement, data modeling, dashboard build, validation, and rollout. Governance includes defining KPI owners, refresh schedules, and exception handling processes. Training and documentation help stakeholders use dashboards correctly and ask better questions.

Implementation roadmap and governance process (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Limitations and ethical considerations Exploratory analysis describes patterns but does not guarantee causal relationships without controlled experiments. Synthetic datasets protect confidentiality but may not capture all operational edge cases of real organizations. Ethical use includes avoiding misuse of customer data and applying appropriate access control in reporting.

Limitations and ethical considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Use of SQL and BI together in industry Industry practice commonly uses SQL for extraction and transformation and BI tools for visualization. Having both layers documented supports auditability because data lineage is more transparent. A data analyst often needs to explain how the same metric is computed in SQL and in BI measures.

Use of SQL and BI together in industry (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Retail pricing strategy and competitive positioning Retailers must balance price competitiveness with margin sustainability, especially when multiple channels exist. Competitive pricing requires monitoring rival prices and understanding customer price sensitivity across categories. An analytics program can set threshold rules to prevent over-discounting while still allowing tactical promotions.

Retail pricing strategy and competitive positioning (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Inventory and assortment considerations Assortment planning depends on understanding category contribution and product-level velocity. High sales categories may still be low margin, so assortment decisions should consider profitability and return behavior. Future expansion can integrate stock on hand and replenishment lead time to connect sales with operations.

Inventory and assortment considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Customer behavior and loyalty Loyalty tiers and customer demographics can influence basket size, discount usage, and satisfaction patterns. High-value customers may prefer consistent service and availability over deep discounts. Segment-based reporting makes promotion planning more efficient because each segment can be targeted differently.

Customer behavior and loyalty (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Data quality risks and controls Missing keys, duplicate transactions, and inconsistent date formats are common risks in retail datasets. Validation rules such as allowed ranges for discounts and delivery days improve trust and prevent misleading charts. Regular reconciliation with finance summaries improves confidence in reported net sales and profit.

Data quality risks and controls (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Visualization principles for business reports Clear hierarchy, minimal clutter, and consistent formatting make dashboards more usable for decision makers. Color can be used to encode meaning such as good vs. bad performance, but it should remain accessible and consistent. Axis labels, units, and titles reduce confusion, especially when dashboards are viewed on different screens.

Visualization principles for business reports (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain

valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Implementation roadmap and governance process A simple roadmap includes requirements gathering, KPI agreement, data modeling, dashboard build, validation, and rollout. Governance includes defining KPI owners, refresh schedules, and exception handling processes. Training and documentation help stakeholders use dashboards correctly and ask better questions.

Implementation roadmap and governance process (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Limitations and ethical considerations Exploratory analysis describes patterns but does not guarantee causal relationships without controlled experiments. Synthetic datasets protect confidentiality but may not capture all operational edge cases of real organizations. Ethical use includes avoiding misuse of customer data and applying appropriate access control in reporting.

Limitations and ethical considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Use of SQL and BI together in industry Industry practice commonly uses SQL for extraction and transformation and BI tools for visualization. Having both layers documented supports

auditability because data lineage is more transparent. A data analyst often needs to explain how the same metric is computed in SQL and in BI measures.

Use of SQL and BI together in industry (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Retail pricing strategy and competitive positioning Retailers must balance price competitiveness with margin sustainability, especially when multiple channels exist. Competitive pricing requires monitoring rival prices and understanding customer price sensitivity across categories. An analytics program can set threshold rules to prevent over-discounting while still allowing tactical promotions.

Retail pricing strategy and competitive positioning (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Inventory and assortment considerations Assortment planning depends on understanding category contribution and product-level velocity. High sales categories may still be low margin, so assortment decisions should consider profitability and return behavior. Future expansion can integrate stock on hand and replenishment lead time to connect sales with operations.

Inventory and assortment considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Customer behavior and loyalty Loyalty tiers and customer demographics can influence basket size, discount usage, and satisfaction patterns. High-value customers may prefer consistent service and availability over deep discounts. Segment-based reporting makes promotion planning more efficient because each segment can be targeted differently.

Customer behavior and loyalty (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Data quality risks and controls Missing keys, duplicate transactions, and inconsistent date formats are common risks in retail datasets. Validation rules such as allowed ranges for discounts and delivery days improve trust and prevent misleading charts. Regular reconciliation with finance summaries improves confidence in reported net sales and profit.

Data quality risks and controls (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Visualization principles for business reports Clear hierarchy, minimal clutter, and consistent formatting make dashboards more usable for decision makers. Color can be used to encode meaning such as good vs. bad performance, but it should remain accessible and consistent. Axis labels, units, and titles reduce confusion, especially when dashboards are viewed on different screens.

Visualization principles for business reports (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Implementation roadmap and governance process A simple roadmap includes requirements gathering, KPI agreement, data modeling, dashboard build, validation, and rollout. Governance includes defining KPI owners, refresh schedules, and exception handling processes. Training and documentation help stakeholders use dashboards correctly and ask better questions.

Implementation roadmap and governance process (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Limitations and ethical considerations Exploratory analysis describes patterns but does not guarantee causal relationships without controlled experiments. Synthetic datasets protect confidentiality but may not capture all operational edge cases of real organizations. Ethical use includes avoiding misuse of customer data and applying appropriate access control in reporting.

Limitations and ethical considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Use of SQL and BI together in industry Industry practice commonly uses SQL for extraction and transformation and BI tools for visualization. Having both layers documented supports auditability because data lineage is more transparent. A data analyst often needs to explain how the same metric is computed in SQL and in BI measures.

Use of SQL and BI together in industry (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Retail pricing strategy and competitive positioning Retailers must balance price competitiveness with margin sustainability, especially when multiple channels exist. Competitive pricing requires monitoring rival prices and understanding customer price sensitivity across categories. An analytics program can set threshold rules to prevent over-discounting while still allowing tactical promotions.

Retail pricing strategy and competitive positioning (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons

remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Inventory and assortment considerations Assortment planning depends on understanding category contribution and product-level velocity. High sales categories may still be low margin, so assortment decisions should consider profitability and return behavior. Future expansion can integrate stock on hand and replenishment lead time to connect sales with operations.

Inventory and assortment considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Customer behavior and loyalty Loyalty tiers and customer demographics can influence basket size, discount usage, and satisfaction patterns. High-value customers may prefer consistent service and availability over deep discounts. Segment-based reporting makes promotion planning more efficient because each segment can be targeted differently.

Customer behavior and loyalty (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Data quality risks and controls Missing keys, duplicate transactions, and inconsistent date formats are common risks in retail datasets. Validation rules such as allowed ranges for discounts

and delivery days improve trust and prevent misleading charts. Regular reconciliation with finance summaries improves confidence in reported net sales and profit.

Data quality risks and controls (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Visualization principles for business reports Clear hierarchy, minimal clutter, and consistent formatting make dashboards more usable for decision makers. Color can be used to encode meaning such as good vs. bad performance, but it should remain accessible and consistent. Axis labels, units, and titles reduce confusion, especially when dashboards are viewed on different screens.

Visualization principles for business reports (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Implementation roadmap and governance process A simple roadmap includes requirements gathering, KPI agreement, data modeling, dashboard build, validation, and rollout. Governance includes defining KPI owners, refresh schedules, and exception handling processes. Training and documentation help stakeholders use dashboards correctly and ask better questions.

Implementation roadmap and governance process (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Limitations and ethical considerations Exploratory analysis describes patterns but does not guarantee causal relationships without controlled experiments. Synthetic datasets protect confidentiality but may not capture all operational edge cases of real organizations. Ethical use includes avoiding misuse of customer data and applying appropriate access control in reporting.

Limitations and ethical considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Use of SQL and BI together in industry Industry practice commonly uses SQL for extraction and transformation and BI tools for visualization. Having both layers documented supports auditability because data lineage is more transparent. A data analyst often needs to explain how the same metric is computed in SQL and in BI measures.

Use of SQL and BI together in industry (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Retail pricing strategy and competitive positioning Retailers must balance price competitiveness with margin sustainability, especially when multiple channels exist. Competitive pricing requires monitoring rival prices and understanding customer price sensitivity across categories. An analytics program can set threshold rules to prevent over-discounting while still allowing tactical promotions.

Retail pricing strategy and competitive positioning (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Inventory and assortment considerations Assortment planning depends on understanding category contribution and product-level velocity. High sales categories may still be low margin, so assortment decisions should consider profitability and return behavior. Future expansion can integrate stock on hand and replenishment lead time to connect sales with operations.

Inventory and assortment considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Customer behavior and loyalty Loyalty tiers and customer demographics can influence basket size, discount usage, and satisfaction patterns. High-value customers may prefer consistent service and availability over deep discounts. Segment-based reporting makes promotion planning more efficient because each segment can be targeted differently.

Customer behavior and loyalty (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Data quality risks and controls Missing keys, duplicate transactions, and inconsistent date formats are common risks in retail datasets. Validation rules such as allowed ranges for discounts and delivery days improve trust and prevent misleading charts. Regular reconciliation with finance summaries improves confidence in reported net sales and profit.

Data quality risks and controls (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Visualization principles for business reports Clear hierarchy, minimal clutter, and consistent formatting make dashboards more usable for decision makers. Color can be used to encode meaning such as good vs. bad performance, but it should remain accessible and consistent. Axis labels, units, and titles reduce confusion, especially when dashboards are viewed on different screens.

Visualization principles for business reports (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain

valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Implementation roadmap and governance process A simple roadmap includes requirements gathering, KPI agreement, data modeling, dashboard build, validation, and rollout. Governance includes defining KPI owners, refresh schedules, and exception handling processes. Training and documentation help stakeholders use dashboards correctly and ask better questions.

Implementation roadmap and governance process (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Limitations and ethical considerations Exploratory analysis describes patterns but does not guarantee causal relationships without controlled experiments. Synthetic datasets protect confidentiality but may not capture all operational edge cases of real organizations. Ethical use includes avoiding misuse of customer data and applying appropriate access control in reporting.

Limitations and ethical considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Use of SQL and BI together in industry Industry practice commonly uses SQL for extraction and transformation and BI tools for visualization. Having both layers documented supports

auditability because data lineage is more transparent. A data analyst often needs to explain how the same metric is computed in SQL and in BI measures.

Use of SQL and BI together in industry (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Retail pricing strategy and competitive positioning Retailers must balance price competitiveness with margin sustainability, especially when multiple channels exist. Competitive pricing requires monitoring rival prices and understanding customer price sensitivity across categories. An analytics program can set threshold rules to prevent over-discounting while still allowing tactical promotions.

Retail pricing strategy and competitive positioning (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Inventory and assortment considerations Assortment planning depends on understanding category contribution and product-level velocity. High sales categories may still be low margin, so assortment decisions should consider profitability and return behavior. Future expansion can integrate stock on hand and replenishment lead time to connect sales with operations.

Inventory and assortment considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Customer behavior and loyalty Loyalty tiers and customer demographics can influence basket size, discount usage, and satisfaction patterns. High-value customers may prefer consistent service and availability over deep discounts. Segment-based reporting makes promotion planning more efficient because each segment can be targeted differently.

Customer behavior and loyalty (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Data quality risks and controls Missing keys, duplicate transactions, and inconsistent date formats are common risks in retail datasets. Validation rules such as allowed ranges for discounts and delivery days improve trust and prevent misleading charts. Regular reconciliation with finance summaries improves confidence in reported net sales and profit.

Data quality risks and controls (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Visualization principles for business reports Clear hierarchy, minimal clutter, and consistent formatting make dashboards more usable for decision makers. Color can be used to encode meaning such as good vs. bad performance, but it should remain accessible and consistent. Axis labels, units, and titles reduce confusion, especially when dashboards are viewed on different screens.

Visualization principles for business reports (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Implementation roadmap and governance process A simple roadmap includes requirements gathering, KPI agreement, data modeling, dashboard build, validation, and rollout. Governance includes defining KPI owners, refresh schedules, and exception handling processes. Training and documentation help stakeholders use dashboards correctly and ask better questions.

Implementation roadmap and governance process (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Limitations and ethical considerations Exploratory analysis describes patterns but does not guarantee causal relationships without controlled experiments. Synthetic datasets protect confidentiality but may not capture all operational edge cases of real organizations. Ethical use includes avoiding misuse of customer data and applying appropriate access control in reporting.

Limitations and ethical considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Use of SQL and BI together in industry Industry practice commonly uses SQL for extraction and transformation and BI tools for visualization. Having both layers documented supports auditability because data lineage is more transparent. A data analyst often needs to explain how the same metric is computed in SQL and in BI measures.

Use of SQL and BI together in industry (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Retail pricing strategy and competitive positioning Retailers must balance price competitiveness with margin sustainability, especially when multiple channels exist. Competitive pricing requires monitoring rival prices and understanding customer price sensitivity across categories. An analytics program can set threshold rules to prevent over-discounting while still allowing tactical promotions.

Retail pricing strategy and competitive positioning (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons

remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Inventory and assortment considerations Assortment planning depends on understanding category contribution and product-level velocity. High sales categories may still be low margin, so assortment decisions should consider profitability and return behavior. Future expansion can integrate stock on hand and replenishment lead time to connect sales with operations.

Inventory and assortment considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Customer behavior and loyalty Loyalty tiers and customer demographics can influence basket size, discount usage, and satisfaction patterns. High-value customers may prefer consistent service and availability over deep discounts. Segment-based reporting makes promotion planning more efficient because each segment can be targeted differently.

Customer behavior and loyalty (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Data quality risks and controls Missing keys, duplicate transactions, and inconsistent date formats are common risks in retail datasets. Validation rules such as allowed ranges for discounts

and delivery days improve trust and prevent misleading charts. Regular reconciliation with finance summaries improves confidence in reported net sales and profit.

Data quality risks and controls (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Visualization principles for business reports Clear hierarchy, minimal clutter, and consistent formatting make dashboards more usable for decision makers. Color can be used to encode meaning such as good vs. bad performance, but it should remain accessible and consistent. Axis labels, units, and titles reduce confusion, especially when dashboards are viewed on different screens.

Visualization principles for business reports (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Implementation roadmap and governance process A simple roadmap includes requirements gathering, KPI agreement, data modeling, dashboard build, validation, and rollout. Governance includes defining KPI owners, refresh schedules, and exception handling processes. Training and documentation help stakeholders use dashboards correctly and ask better questions.

Implementation roadmap and governance process (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Limitations and ethical considerations Exploratory analysis describes patterns but does not guarantee causal relationships without controlled experiments. Synthetic datasets protect confidentiality but may not capture all operational edge cases of real organizations. Ethical use includes avoiding misuse of customer data and applying appropriate access control in reporting.

Limitations and ethical considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Use of SQL and BI together in industry Industry practice commonly uses SQL for extraction and transformation and BI tools for visualization. Having both layers documented supports auditability because data lineage is more transparent. A data analyst often needs to explain how the same metric is computed in SQL and in BI measures.

Use of SQL and BI together in industry (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Retail pricing strategy and competitive positioning Retailers must balance price competitiveness with margin sustainability, especially when multiple channels exist. Competitive pricing requires monitoring rival prices and understanding customer price sensitivity across categories. An analytics program can set threshold rules to prevent over-discounting while still allowing tactical promotions.

Retail pricing strategy and competitive positioning (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Inventory and assortment considerations Assortment planning depends on understanding category contribution and product-level velocity. High sales categories may still be low margin, so assortment decisions should consider profitability and return behavior. Future expansion can integrate stock on hand and replenishment lead time to connect sales with operations.

Inventory and assortment considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Customer behavior and loyalty Loyalty tiers and customer demographics can influence basket size, discount usage, and satisfaction patterns. High-value customers may prefer consistent service and availability over deep discounts. Segment-based reporting makes promotion planning more efficient because each segment can be targeted differently.

Customer behavior and loyalty (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Data quality risks and controls Missing keys, duplicate transactions, and inconsistent date formats are common risks in retail datasets. Validation rules such as allowed ranges for discounts and delivery days improve trust and prevent misleading charts. Regular reconciliation with finance summaries improves confidence in reported net sales and profit.

Data quality risks and controls (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Visualization principles for business reports Clear hierarchy, minimal clutter, and consistent formatting make dashboards more usable for decision makers. Color can be used to encode meaning such as good vs. bad performance, but it should remain accessible and consistent. Axis labels, units, and titles reduce confusion, especially when dashboards are viewed on different screens.

Visualization principles for business reports (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain

valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Implementation roadmap and governance process A simple roadmap includes requirements gathering, KPI agreement, data modeling, dashboard build, validation, and rollout. Governance includes defining KPI owners, refresh schedules, and exception handling processes. Training and documentation help stakeholders use dashboards correctly and ask better questions.

Implementation roadmap and governance process (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Limitations and ethical considerations Exploratory analysis describes patterns but does not guarantee causal relationships without controlled experiments. Synthetic datasets protect confidentiality but may not capture all operational edge cases of real organizations. Ethical use includes avoiding misuse of customer data and applying appropriate access control in reporting.

Limitations and ethical considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Use of SQL and BI together in industry Industry practice commonly uses SQL for extraction and transformation and BI tools for visualization. Having both layers documented supports

auditability because data lineage is more transparent. A data analyst often needs to explain how the same metric is computed in SQL and in BI measures.

Use of SQL and BI together in industry (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Retail pricing strategy and competitive positioning Retailers must balance price competitiveness with margin sustainability, especially when multiple channels exist. Competitive pricing requires monitoring rival prices and understanding customer price sensitivity across categories. An analytics program can set threshold rules to prevent over-discounting while still allowing tactical promotions.

Retail pricing strategy and competitive positioning (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Inventory and assortment considerations Assortment planning depends on understanding category contribution and product-level velocity. High sales categories may still be low margin, so assortment decisions should consider profitability and return behavior. Future expansion can integrate stock on hand and replenishment lead time to connect sales with operations.

Inventory and assortment considerations (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

Customer behavior and loyalty Loyalty tiers and customer demographics can influence basket size, discount usage, and satisfaction patterns. High-value customers may prefer consistent service and availability over deep discounts. Segment-based reporting makes promotion planning more efficient because each segment can be targeted differently.

Customer behavior and loyalty (extended discussion) This section further elaborates practical considerations, including how stakeholders interpret metrics and what operational decisions can follow. The same KPI definitions should be maintained so comparisons remain valid across weeks, cities, and categories. Documentation and validation steps reduce confusion when the dashboard is shared with multiple teams.

5.11 Additional Discussion and Managerial Recommendations (Expanded)

Managerial recommendations are provided to convert the analytical findings into practical actions that a retail leadership team can implement. The recommendations are aligned with the KPI framework used in this project so that each action can be tracked through measurable outcomes such as net sales, profit, margin, return rate, AOV, and CSAT.

1) Discount Guardrails and Approval Workflow: The analysis indicates that discounting improves short-term sales but can reduce profitability if applied broadly without control. RoyalMart should introduce category-wise and channel-wise discount guardrails. If a discount

exceeds a defined threshold, an approval workflow should be used (for example Store Manager → Category Head). This reduces margin leakage while still allowing promotional flexibility during seasonal campaigns. KPIs to track include Profit Margin (%), Profit (SAR), Avg Discount (%), and Net Sales (SAR).

2) Weekly Category Scorecards: RoyalMart should institutionalize a weekly category scorecard to review category performance on Net Sales, Profit, Margin, Avg Discount, CSAT, and Return Rate. Categories with high sales but weak margins should be reviewed for vendor terms, product mix, and campaign design. Categories with high returns should trigger product quality and process reviews. This increases accountability and supports systematic improvement.

3) Online Service-Level Targets Linked to CSAT: For online orders, customer satisfaction is strongly influenced by delivery performance. Since delivery days are captured for online orders, RoyalMart can define service-level targets and monitor their impact on CSAT. A practical approach is to identify the delivery-day threshold after which CSAT declines and then set operational targets to stay within that threshold for high-volume cities. KPIs to track include Avg Delivery Days, Avg CSAT, and Online Return Rate.

4) Returns Root-Cause Program: Returns should be managed as a profitability and experience problem, not only as a refund process. Return reasons should be prioritized by frequency and refund amount. “Damaged” and “Defective” reasons require upstream fixes such as packaging standards and supplier quality checks, while “Wrong item” requires picking accuracy improvements and barcode checks. A monthly Returns Root-Cause Review should assign owners and deadlines, and progress should be tracked in the next dashboard review cycle.

5) Data Governance and KPI Dictionary: To avoid inconsistent reporting, RoyalMart should maintain a KPI dictionary defining each KPI, its formula, source columns, refresh schedule, and

validation checks. Before publishing dashboards, a standard SQL validation checklist should reconcile totals (for example month-wise net sales and profit) and test data quality rules (discount range, delivery days range, null key checks). This improves trust and reduces disputes in business reviews.

6) Dashboard Adoption and Communication: RoyalMart should create a short dashboard user guide explaining what each page answers, how filters work, and how to interpret KPI changes. Brief training for stakeholders increases adoption and ensures dashboards lead to action rather than only reporting.

Overall, these recommendations create a closed-loop system: dashboards support monitoring, drill-down supports diagnosis, corrective actions are implemented with owners and deadlines, and impact is measured using the same KPI framework in subsequent review cycles.

Chapter 6: Conclusion and Future Scope

This chapter summarizes the project outcomes and highlights how the complete workflow demonstrates the role of a data analyst in a retail environment. Using the RoyalMart dataset, the project established a KPI framework, validated calculations through MySQL-style queries, and presented insights through systematic Power BI dashboards with evidence screenshots embedded at relevant analysis points.

Conclusion: The project shows that a transaction-level dataset can be transformed into decision-ready information when analysis is structured, validated, and communicated clearly. The KPI results provide a financial baseline through net sales and profit, while margin analysis highlights the importance of discount management. Customer experience KPIs such as CSAT and return rate provide early warnings about operational problems that can affect long-term customer value. The integration of SQL and Power BI strengthens credibility because metrics are validated and then communicated visually for fast decision support.

Key contributions of this project include (a) defining repeatable KPIs with consistent formulas, (b) applying a star-schema style data model suitable for BI reporting, (c) building executive and diagnostic dashboards, and (d) documenting evidence for SQL execution and dashboard views to support auditability.

Future Scope: The current project focuses on descriptive and exploratory analytics. In future work, RoyalMart can extend the study to predictive analytics by forecasting sales at category and city level using time series methods (moving averages, ARIMA/Prophet) or machine-learning regression models with seasonal and promotional features. Forecasting can improve inventory planning, staffing, and marketing budget allocation.

Customer segmentation can be introduced using RFM (Recency, Frequency, Monetary) analysis and clustering to identify high-value customers, discount-sensitive customers, and churn-risk groups. This enables targeted promotions that protect margin while improving retention and satisfaction.

Another extension is to build an automated data pipeline. Instead of manual CSV updates, data can be loaded into a database or data warehouse, refreshed on a schedule, and validated with automated data quality checks. Alerts can be generated for anomalies such as sudden return spikes or margin drops.

Finally, experimentation can be introduced to establish causality. For example, A/B testing can be used to evaluate new discount levels, delivery SLA changes, and packaging improvements. This helps confirm which actions truly improve CSAT and reduce returns, enabling evidence-based investment decisions.

In summary, the project provides a strong foundation for retail analytics. With forecasting, segmentation, automation, and experimentation, RoyalMart can evolve from descriptive reporting to a mature decision-intelligence system that continuously improves profitability and customer experience.

References

Kimball, R., & Ross, M. (2013). The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling (3rd ed.). Wiley.

Provost, F., & Fawcett, T. (2013). Data Science for Business. O'Reilly Media.

Sharda, R., Delen, D., & Turban, E. (2019). Business Intelligence, Analytics, and Data Science: A Managerial Perspective (4th ed.). Pearson.

CRISP-DM Consortium. (2000). CRISP-DM 1.0: Step-by-step data mining guide.

Microsoft. (n.d.). Power BI documentation. Microsoft Learn.

Appendix A: Data Dictionary Summary

A detailed data dictionary is included in the dataset package as DATA_DICTIONARY.md. The most important fields for analysis are channel, category, discount_pct, net_sales, profit, delivery_days, csat_score, return_reason, and refund_amount.

Appendix B: Power BI DAX Measures

Note: These measures assume table names 'transactions' and 'returns'. Replace names if your Power BI model uses different names.

Net Sales (SAR)

```
Net Sales (SAR) = SUM ( transactions[net_sales] )
```

Profit (SAR)

```
Profit (SAR) = SUM ( transactions[profit] )
```

Profit Margin (%)

```
Profit Margin (%) = DIVIDE ( [Profit (SAR)], [Net Sales (SAR)] ) * 100
```

Orders

```
Orders = DISTINCTCOUNT ( transactions[transaction_id] )
```

AOV (SAR)

```
AOV (SAR) = DIVIDE ( [Net Sales (SAR)], [Orders] )
```

Avg Discount (%)

```
Avg Discount (%) = AVERAGE ( transactions[discount_pct] )
```

Avg CSAT

```
Avg CSAT = AVERAGE ( transactions[csat_score] )
```

Returns Count

```
Returns Count = COUNTROWS ( returns )
```

Refund Amount (SAR)

```
Refund Amount (SAR) = SUM ( returns[refund_amount] )
```

Return Rate (%)

```
Return Rate (%) = DIVIDE ( [Returns Count], [Orders] ) * 100
```